

< DPM Solver >

Reverse process "SDE" follows $dx(t) = \left[f(t, x(t)) + \frac{g^2(t)}{1 - \hat{E}_0} \hat{E}_0(x(t), t) \right] dt + g(t) dW_t$
 $= \frac{g^2(t)}{1 - \hat{E}_0} \hat{E}_0(x(t), t)$

→ $\frac{dx(t)}{dt} = f(t, x(t)) + \frac{g^2(t)}{2\sigma t} \hat{E}_0(x(t), t)$ is the corresponding "ODE" by Ito-Stratonovich equation

Let's derive $f(t, x)$, $g(t, x)$ from forward process "ODE"

SDE forward process first

$$\rightarrow dx(t) = f(t, x(t))dt + g(t)dW_t = N(dx(t)z_0, \sigma^2(t)I)$$

→ we newly define the model which satisfy ...

$$\left\{ \begin{array}{l} \textcircled{1} q(x_t|z_t) = N(dx(t)z_0, \sigma^2(t)I) \\ \textcircled{2} \text{SNR}(\text{signal} \rightarrow \text{noise ratio}) \frac{dx^2(t)}{\sigma^2(t)} \text{ strictly decrease} \end{array} \right\} \text{ as 'Diffusion Model'}$$

Let's start from expressing $f(t, x)$, $g(t, x)$ using $x(t)$, $\sigma(t)$

< $f(t, x)$ > → by comparing the 'mean' value

from forward SDE process...

$$dx(t) = f(t, x(t))dt + g(t)dW_t$$

$$\rightarrow E[dx(t)] = E[f(t, x(t))dt + g(t)dW_t] \\ = f(t)E[x(t)]dt + E[g(t)dW_t]$$

→ non-deterministic brownian motion ... which follows $E[W] = 0$

$$\rightarrow \frac{E[dx(t)]}{dt} = f(t)E[x(t)]$$

$$\rightarrow \frac{E[x(t+\Delta t) - x(t)]}{\Delta t} = f(t)E[x(t)]$$

$$\rightarrow \frac{d}{dt} E[x(t)] = f(t)E[x(t)]$$

$$\rightarrow \frac{\frac{d}{dt} E[x(t)]}{E[x(t)]} = f(t)$$

$$\rightarrow \frac{d}{dt} \left(\log E[x(t)] - \log E[x(0)] \right) = f(t)$$

$$\rightarrow \log E[x(t)] = \log E[x(0)] + \int_0^t f(k)dk$$

$$\rightarrow E[x(t)] = E[x(0)] e^{\int_0^t f(k)dk} \quad \text{since } x_t \text{ can be given when } x_0 \text{ exists,}$$

$$E[x_1|z_0] = x_0 e^{\int_0^t f(k)dk}$$

from forward process of Diffusion model ...

$$p(x_1|z_0) = N(dx(t)z_0, \sigma^2(t)I), \quad E[x_1|z_0] = dx(t)z_0$$

$$dx(t) = e^{\int_0^t f(k)dk}$$

$$f(t) = \frac{d}{dt} \log dx(t)$$

$\langle g(x) \rangle \rightarrow$ by comparison 'variance' using $d\langle x^2 \rangle$

let $\Delta(x) = \frac{1}{\Delta(x)}$

$$\textcircled{1} d\langle x^2 \rangle = (\Delta + d\Delta)(x + dx) - \Delta x$$

$$= \Delta dx + x d\Delta + d\Delta x$$

$\rightarrow d\Delta$ is deterministic.
dx crosses at brown noise $\uparrow$ by Ito calculus.

$\textcircled{2}$ from forward SDE process..

$$dx(t) = f(x,t)dt + g(x,t)dW_t$$

$$\overset{x\Delta}{\rightarrow} \Delta dx = \Delta f x dt + \Delta g dW$$

recall that $\frac{d}{dt} \Delta(x) = -\Delta(x)f(x)$..

$$\rightarrow d\Delta = -\Delta f dt$$

with $\textcircled{2}$, $\Delta dx = -x d\Delta + \Delta g dW$

with $\textcircled{1}$, $d\langle x^2 \rangle = \Delta g dW$

$$\rightarrow \int_0^t \Delta g(x(s)) dW_s = \int_0^t \Delta(x(s)) g(x(s)) dW_s$$

$$\rightarrow \Delta(x(t)) x(t) - \Delta(x(0)) x(0) = \int_0^t \frac{\partial \Delta(x)}{\partial x} dW_s$$

$$\rightarrow x(t) = \frac{\partial \Delta(x)}{\partial x} x(t) + \Delta(x) \int_0^t \frac{\partial g(x)}{\partial x} dW_s$$

$$= \Delta(x(t)) + \Delta(x(0)) \int_0^t \frac{\partial g(x)}{\partial x} dW_s$$

$$\rightarrow x(t) - \Delta(x(t)) x(t) = \Delta(x(0)) \int_0^t \frac{\partial g(x)}{\partial x} dW_s$$

$= E[x(t)|x(0)]$

$$\rightarrow V_\Delta[x(t)|x(0)] = E[(x(t) - E[x(t)|x(0)])^2] = E\left[\Delta^2(x(0)) \left(\int_0^t \frac{\partial g(x)}{\partial x} dW_s\right)^2\right]$$

$$= \Delta^2(x(0)) \int_0^t \frac{\partial g(x)^2}{\partial x^2} dt$$

from diffusion process ..

$$\sigma^2(t) = \Delta^2(t) \int_0^t \frac{\partial g(x)^2}{\partial x^2} dt$$

$$\text{let } \int_0^t \frac{\partial g(x)^2}{\partial x^2} dt = I(t)$$

$$\frac{d}{dt} \sigma^2(t) = 2\Delta(t) \Delta'(t) I(t) + \Delta(t)^2$$

$$\rightarrow \text{use } \frac{d}{dt} \Delta(t) = -\Delta(t)f(t)$$

$$= 2\Delta(t) \cdot \Delta'(t) I(t) + \Delta(t)^2$$

$$= 2f(t) \Delta(t)^2 I(t) + \Delta(t)^2$$

$= \sigma^2(t)$

$$\frac{d}{dt} \sigma^2(t) = 2f(t) \sigma^2(t) + \Delta(t)^2$$

$$\rightarrow \sigma^2(t) = \frac{d\Delta(t)}{dt} - 2f(t)\Delta(t)$$

$$= \frac{d\Delta(t)}{dt} - 2\left(\frac{d}{dt} \log \Delta(t)\right) \sigma^2(t)$$

$$\begin{aligned} & \frac{d}{dt} \sigma^2(t)^2 - 2\left(\frac{d}{dt} \log \Delta(t)\right) \sigma^2(t)^2 \\ &= 2\sigma^2(t) \cdot \frac{d}{dt} \sigma^2(t) - 2\sigma^2(t)^2 \cdot \frac{d}{dt} \log \Delta(t) \\ & \quad \rightarrow \sigma^2(t) \cdot \frac{d}{dt} \log \sigma^2(t) \\ &= 2\sigma^2(t)^2 \left(\frac{d}{dt} \log \sigma^2(t) - \frac{d}{dt} \log \Delta(t) \right) \\ &= 2\sigma^2(t)^2 \left(-\frac{d}{dt} \lambda(t) \right), \quad \lambda(t) = \log \frac{\Delta(t)}{\sigma^2(t)} \end{aligned}$$

$$\Rightarrow \sigma^2(t)^2 = 2\sigma^2(t)^2 \left(-\frac{d\lambda}{dt} \right)$$

Back to reverse ODE process

$$\frac{dx(s)}{ds} = \lambda(s)x(s) + \frac{p(s)}{2s\lambda(s)} \hat{e}_0(z, s)$$

Let's plug $\lambda(s), x(s)$

$$\rightarrow \frac{dx(s)}{ds} = \left(\frac{d\lambda(s)}{ds} \right) x(s) - \lambda(s) \frac{d\lambda(s)}{ds} \hat{e}_0$$

↳ Same as probability law deriv...

⇒ Anyway, the eqn above is 1st-order semilinear differential equation

< 1st-order semilinear differential eq. >

$$x'(s) = p(s)x(s) + q(x(s), s)$$

$$\rightarrow x'(s) - p(s)x(s) = q(x(s), s)$$

$$\times e^{-\int_s^t p(\tau) d\tau} \rightarrow x(s) e^{-\int_s^t p(\tau) d\tau} - p(s)x(s) e^{-\int_s^t p(\tau) d\tau} = q(x(s), s) e^{-\int_s^t p(\tau) d\tau}$$

$$\text{note that } \left(e^{-\int_s^t p(\tau) d\tau} \right)' = -p(s) e^{-\int_s^t p(\tau) d\tau}$$

$$\rightarrow \frac{d}{ds} \left(x(s) e^{-\int_s^t p(\tau) d\tau} \right) = q(x(s), s) e^{-\int_s^t p(\tau) d\tau}$$

$$\rightarrow x(s) e^{-\int_s^t p(\tau) d\tau} = x(s) e^{-\int_s^t p(\tau) d\tau} + \int_s^t q(x(\tau), \tau) e^{-\int_s^t p(\tau) d\tau} d\tau$$

$$\Rightarrow x(s) = x(s) e^{\int_s^t p(\tau) d\tau} + \int_s^t q(x(\tau), \tau) e^{\int_s^t p(\tau) d\tau} d\tau$$

Back to reverse ode process...

$$\frac{dx(s)}{ds} = \left(\frac{d\lambda(s)}{ds} \right) x(s) - \lambda(s) \frac{d\lambda(s)}{ds} \hat{e}_0 \quad \text{Can be seen as...}$$

$$x'(s) = p(s)x(s) - q(x(s), s)$$

$$\Rightarrow \text{plug } p(s), q(s) \text{ into } x(s) = x(s) e^{\int_s^t p(\tau) d\tau} + \int_s^t q(x(\tau), \tau) e^{\int_s^t p(\tau) d\tau} d\tau$$

$$x(s) = \frac{d\lambda}{ds} x(s) - d\lambda \int_s^t \left(\frac{d\lambda_\tau}{d\lambda_\tau} \right) e^{-\lambda_\tau} \hat{e}_0(x(\tau), \lambda_\tau) d\lambda_\tau$$

$$= \frac{d\lambda}{ds} x(s) - d\lambda \int_{\lambda_s}^{\lambda_t} e^{-\lambda_\tau} \hat{e}_0(x(\tau), \lambda_\tau) d\lambda_\tau$$

< DPM Solver basic eqn >

Why $e^{-\int p ds}$?

→ in linear ODE, $x'(s) = p(s)x(s)$

$$\rightarrow \frac{x'(s)}{x(s)} = p(s)$$

$$\rightarrow \ln x(s) = \int p(s)$$

$$\rightarrow x(s) = e^{\int p(s)}$$

$$\frac{d}{ds} \left(e^{\int p(s)} \right) = p(s) e^{\int p(s)}$$

$$x(t) = \frac{dt}{ds} x(s) - dt \int_{\lambda_s}^{\lambda_t} e^{-\lambda t} \hat{E}_0(x(\lambda_t), \lambda_t) d\lambda_t$$

< DPM - Solver - 1 >

$\hat{E}_0$ is irreducible, so let's assume that $\hat{E}_0$ is a constant in a time step $(\lambda_t, \lambda_{t+1})$ in discrete step.
↳ at the end

$$\begin{aligned} x(t) &= \frac{dt}{dt+1} x(t_n) - dt \hat{E}_0 \int_{\lambda_{t+1}}^{\lambda_t} e^{-\lambda} d\lambda \\ &= \frac{dt}{dt+1} x_{t+1} - dt \underbrace{(e^{-\lambda_t} + e^{-\lambda_{t+1}})} \hat{E}_0(\lambda_{t+1}) \end{aligned}$$

$$\hookrightarrow dt \left(-\frac{e^{-\lambda_t}}{\lambda_t} + \frac{e^{-\lambda_{t+1}}}{\lambda_{t+1}} \right)$$

$$= dt \frac{e^{-\lambda_{t+1}}}{\lambda_{t+1}} - e^{-\lambda_t}$$

$$= e^{-\lambda_t} \left(\frac{1 - e^{-\lambda_t}}{\lambda_t} - 1 \right)$$

$$= e^{-\lambda_t} \left(e^{\lambda_t - \lambda_{t+1}} - 1 \right)$$

$$\therefore x(t) = \frac{dt}{dt+1} x_{t+1} - e^{-\lambda_t} (e^{\lambda_t - \lambda_{t+1}} - 1) \hat{E}_0(\lambda_{t+1})$$

We can use Runge Kutta methods for higher order !!

